# **Tour Ceylon:** AI-Based Tourism Safety and Itinerary Platform

Team Presentation — Progress Presentation 2 (PP2)

- R26-IT-152 | Final Year Project (PP2) — SLIIT, Dept. of Information Technology
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# Problem & Research Gap

## Fragmented Ecosystem

Sri Lanka's tourism sector runs on disconnected tools: no unified path from tourist distress to dispatchable police coordination, no data-grounded trip planning beyond generic AI that can invent places, no way to match tourists to verified wellness centers among 83+ options, and no site-level crowd or risk forecasting across the country's tourist destinations.

## Reactive, Not Predictive

Existing platforms operate in silos — responding to incidents after they occur rather than anticipating risk before it materializes.

## Identified Research Gap

No platform in Sri Lanka currently unifies location-aware emergency dispatch, rule-based (non-hallucinating) itinerary planning, dosha-aware wellness matching, and ML-driven crowd forecasting into a single system — each of these problems has been solved in isolation elsewhere, never together, and never for Sri Lanka



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# System Architecture

A **4-component monorepo** unified through a shared Flask backend and PostgreSQL database — each component independently developed yet fully integrated.

### Crowd & Risk Prediction

Random Forest model across **197 tourist sites** — site-level risk scoring and crowd density forecasting

### SOS Emergency Response

GPS-based, AI-triaged incident routing to police, hospitals, and embassies in real time

### Itinerary Planning

6-step wizard with rule-based scheduling across moods, categories, and tourist preferences

### Wellness & Ayurveda

Dosha classifier + NLP review scoring for personalized wellness matching

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# Crowd & Risk Prediction

1

## Dataset Scale

**197** tourist sites · **144,007** records

2

## Model: Random Forest

`max_depth=12` · `min_samples_split=7` · `n_estimators=100`

3

## Output

Risk scoring + personalized site ranking by crowd density and safety metrics



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# SOS Emergency Response Component



### Incident Capture

One-tap SOS with voice confirmation, GPS capture, and **16 incident types** supported



### AI-Assisted Dispatch

0–100 risk scoring with auto-routing to nearest of **695 police stations**, **150+ hospitals**, and **50+ embassies**



### Real-Time Updates

Socket.IO with HTTP polling fallback; offline queue with auto-sync ensures no lost incidents



### Security & Access

RBAC + MFA for officers; **11 tourist languages** supported across the interface



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# Itinerary Planning & Wellness Components

## Itinerary Planning

6-step planning wizard with rule-based scheduling, PDF export, and **10 UI languages**

## Wellness & Ayurveda Matching

Dual-path wizard with dosha classifier and NLP-based review scoring

**12** Ayurveda conditions · **6** retreat types

**83** wellness centers · **256** curated reviews

Dosha classifier validated at **81% accuracy**

**130** attractions · **944** accommodations · **126** businesses

**417** agencies · **2,906** guides

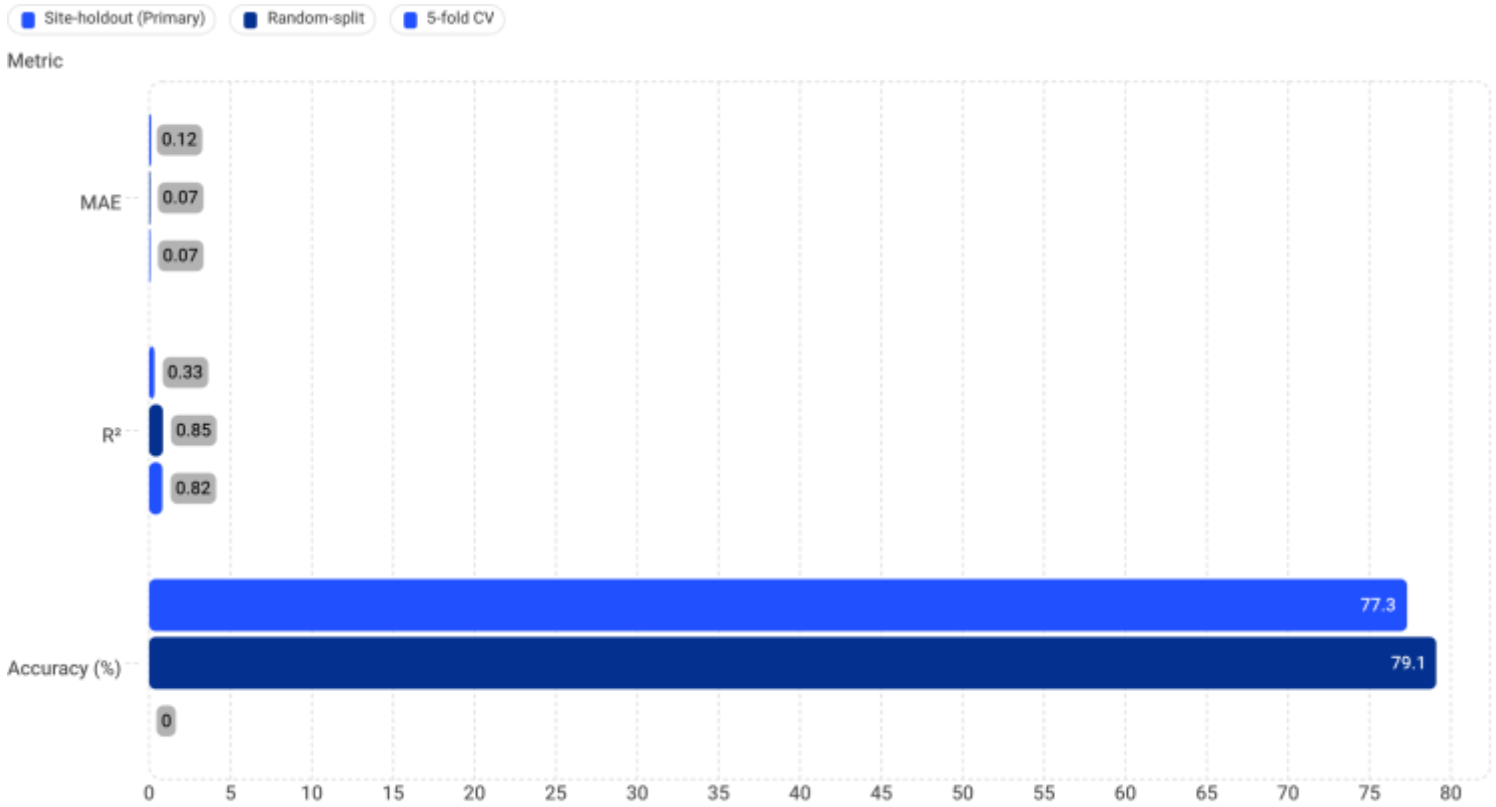
**10 moods** / **6 categories** for personalized filtering



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# Evaluation Results: Crowd Prediction Model

Site-holdout is the **primary metric** — it tests performance on entirely unseen locations, not familiar rows. This reflects honest generalization.




5-fold CV (MAE **0.0694 ± 0.0090**, R² **0.8194 ± 0.0218**) confirms model stability. The lower site-holdout R² (0.3331) reflects the genuine difficulty of generalizing to *unseen* tourist locations — an honest and rigorous evaluation design.

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# Integration Status & Testing

### Live-tested across 4 frontends

All components verified on a shared backend running simultaneously with real seeded data.

### Component verification

**Crowd/Risk:** Fully verified end-to-end · **Itinerary & Wellness:** Verified with live data · **SOS:** Tourist flows, police dashboard, and overstay-detection logic all confirmed functional.

### Resolved gap — transparent disclosure

A missing backend route (`reference/safety-card` endpoint) was identified through systematic audit, fixed, and re-verified — demonstrating rigorous integration methodology.

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# Non-Functional Requirements & Rigor

### Performance

<2s response time for all critical operations · 99.5% uptime target

### Security

Encrypted PII · SHA256 evidence checksums (SOS) · RBAC + MFA for officers

### Accessibility

WCAG 2.1 AA target · multi languages across all platform components

### Academic Transparency

**11 real bugs** were identified through systematic audits across the integrated platform. Bugs were fixed where feasible; remaining limitations are **transparently disclosed** rather than concealed — reflecting honest, rigorous academic methodology.

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# Commercialization & Scalability

## Applicable Stakeholders

- Sri Lanka Tourism Development Authority
- Hotel chains & travel agencies
- Insurance & safety partners
- National Tourist Police

## Use Cases & Scalability

- Occupancy forecasting for site management
- Itinerary automation at national scale
- Risk-based tourist safety advisories
- Emergency response modernization

Architecture is extensible to other destinations with comparable site-level tourism data.



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# Thank You

**Tour Ceylon — SafeJourney AI (R26-IT-152)**

Team Presentation — PP2


### Research Team

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